Distillation

Pre-Lab Plan Unit Operations Lab # 4

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Group 3, Tuesday, PM section

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1. Experimental Objective (5 pts)

This lab has two distinct objectives which demonstrate the effect of changing different parameters on the overall performance of the distillation column. In part A, the objective is to quantify the boil-up rate and pressure drop over the column as a function of reboiler power. Additionally, these quantities and the top/bottom liquid compositions are used to understand the efficiency of the system as a function of reboiler power. In part B, top and bottom compositions are varied over time as a function of reflux ratio (which is regulated through a controller).

2. Experimental

The distillation column apparatus (shown below in **Figure 1**) is utilized for both experiments A and B. Initially, 10L of the water-ethanol mixture that will be separated during the experiments can be added to the drum (#**1**) and heated by turning on the power through the safe area console (**#9**). This controller manages several key parameters for the column, including reboiler power (varied in experiment A) and the reflux rate/valve (**#3 -** used to vary reflux ratio in experiment B). Additionally, the console and computer can be used to read temperature values from thermocouples throughout the system. Once turned on, the solution matriculates through the system and separation occurs across the distillation column (**#2**).

Another key component of both experiments is understanding the composition of the top and bottom products of the column. Samples of each product can be collected from the top and bottom product receivers (**#6** and **#7**).

2.1 Apparatus (5 pts)

Figure 1: Apparatus with color coded labels – picture of actual apparatus

Figure 2: Apparatus with color coded labels – picture of schematic.

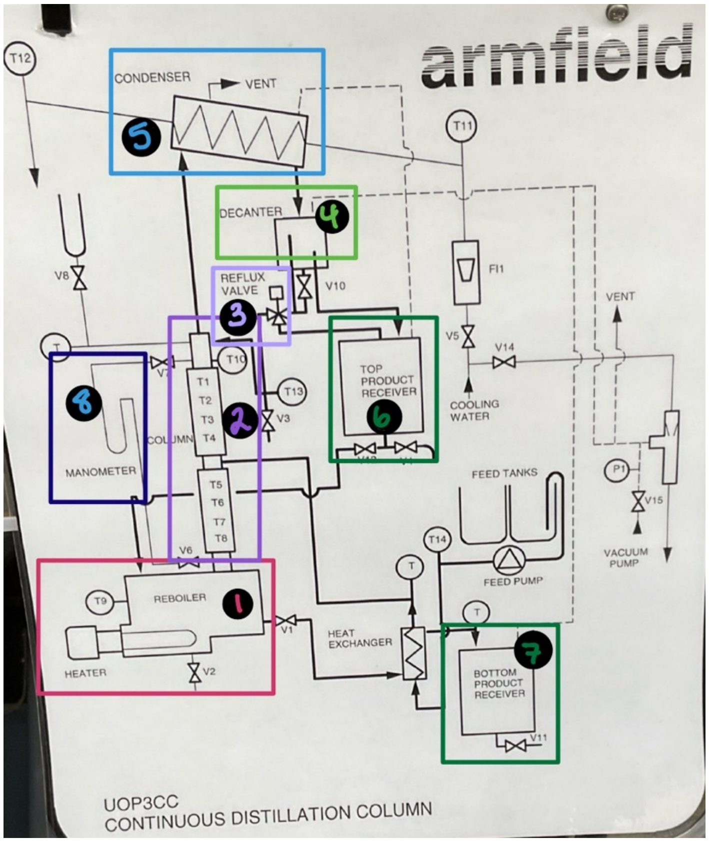

Table 1: Relates the color-coded labels to what part of the apparatus they are.

Number/Color	Apparatus Part
1 – Pink	Reboiler w/ heater
2 – Purple	Distillation Column
3 – Light Purple	Reflux Valve
4 – Light Green	Decanter
5 – Light Blue	Condenser
6 – Dark Green	Top Product Receiver
7 – Dark Green	Bottom Product Receiver
8 – Dark Blue	Manometer
9 – Dark Blue	Safe Area Console

Apparatus parts eight and nine in dark blue relate to specific reading equipment. Apparatus parts six and seven in dark green relate to product receiving. Apparatus part in one in pink boils the ethanol/water mixture, to then go through apparatus part two in purple – the reflux of the vapor is controlled by apparatus part three in light purple. The vapor product is condensed with apparatus part five in light blue. The decanter is apparatus part four in light green.

2.2 Materials and Supplies (5 pts)

Table 2: A list of the materials/supplies used for lab.

Material/Supplies	**Purpose/Specifications**
200 Proof Ethanol	For the ethanol/water mixture that is being distilled
DI Water	For the ethanol/water mixture that is being distilled
Cooling Water	Used for the condenser
Funnel	To get the ethanol/water mixture into the reboiler
Graduated Cylinder	To make the ethanol/water mixture
Beaker	For various product collection, and making the ethanol/water mixture
Test Tube	For product collection, and testing
Parafilm	To avoid evaporation of hot product – to keep the ethanol/water composition from changing of the product as the product cools off
Portable Alcohol Reader	To measure the composition of the product
Computer	To collect data from the Safe Area Console

2.3 Experimental Plan (10 pts)

- Create 10L of a 10 wt% solution with 200 proof ethanol in the fume hood.
- Ensure all valves are closed and fill the reboiler with the prepared solution.
- For experiment A, turn on the Safe Area Console and set the temperature in the reboiler to T9.
- Introduce the cooling water and adjust the flow until Fl1 = ~3L/min.
- Set the reboiler power to 1.50kW and power on the reboiler.
- Open valves V6 and V7 to allow vapor circulation, then close the valves.
- Once steady bubbling is observed over all trays, reduce the boiler power to 0.75kW.
- Wait until temperatures T1-T8 are stable, indicating equilibrium.
- Open V3 and collect condensate for 1 minute to measure boil up ratio.
- Once samples have been collected, measure the pressure drop over the column by opening V6 and V7 and then close the valves once measurements have been recorded.
- Cover the samples with parafilm to avoid evaporation of the samples.
- Place test tubes in a cooling bath apparatus.
- Using the Snap 41 Densitometer, measure the volume and composition in each sample.
- Repeat this process for bottoms sample collection, using the same timed-collection and volume measurement procedure.
- Repeat the procedure over 3 trials for both the top and bottom product.
- Record temperatures at T1-T8 and calculate the average temperature of the column.
- For experiment B, use the same solution that is already inside the reboiler.
- Maintain the reboiler power at 0.75kW to ensure steady bubbling over the trays.
- Allow the system to reach equilibrium (about 15 minutes).
- Record temperatures from T1-T8 once they are stable.
- Set the reflux ratio to 25:5 using the reflux valve.

- Every 5 minutes, measure the boil up using the same timed collection procedure as above.
- Collect the top and bottoms samples at each timed interval.
- Use the Snap 41 Densitometer to measure the volume and composition.
- After completing data collection at R=25:5, adjust the reflux ratio to 50:5 and repeat the steps above.
- Repeat the process for 3 trials at each reflux ratio to verify consistency.
- Record and average the tray temperatures for each trial and reflux ratio.

The independent variables in this experiment include the reboiler power, reflux ratio and time while the dependent variables are the boil up rate, column pressure drop, ethanol composition in the top and bottoms samples and the average column temperature. To calibrate Densitometer, use DI water and an ethanol standard before each set of measurements. For flow measurements, use the same container for each trial. To calibrate the thermocouples, verify one sensor against a calibrated thermometer at both ambient and 70℃ water-bath conditions. Check for trapped vapor and consistent fluid levels before each reading.

Each experiment condition will be performed in a triplicate and the mean and a 95% confidence interval will be reported using:

$$\bar{x} \pm t_{0.975,n-1}\frac{s}{\sqrt{n}}$$

Where n=3.

The goodness of fit for regression plots will be assessed via R^2 and RMSE values. Outliers exceeding one standard deviation will be re-tested when time permits.

Data from experiment A will be analyzed using the Fenske equation to determine the number of theoretical stages at total reflux, and the corresponding column efficiency will be calculated. A log-log scale will be used to plot ΔP and the boil up rate in order to identify flow-regime transitions. Data from experiment B will be used to construct a McCabe-Thiele Diagram based on measured top and bottoms compositions at reach reflux ratio. The experimental points will be compared to theoretical equilibrium stages to estimate the Murphree tray efficiency.

Project Safety Evaluation (10 pts)

This experiment involves operating a distillation column to separate a 10 wt% ethanol-water mixture under both total and constant reflux conditions. There are several safety concerns including handling high-proof flammable ethanol, managing hot surfaces and pressurized vapor and operating cooling and electrical systems.

200-proof ethanol is a highly flammable liquid and vapor. Exposure to an open flame, hot surfaces or electrical sparks could result in ignition. Vapors may accumulate and travel to ignition sources. In order to control this safety concern, all solutions will be prepared in a fume hood and all ignition sources will be kept away from the solution. All samples will also be covered and sealed with parafilm to prevent vapor loss and accumulation. Although not hazardous, water can cause burns or splashes when mixed with hot ethanol so to avoid any potential risks, all condensate and bottoms samples will be handled carefully to avoid scalding.

The reboiler operates at a high temperature with a maximum power setting of 1.5kW which generates vapor pressures slightly above atmospheric. The hot surfaces on the reboiler and valves pose potential burn risks. To reduce the risk, heat-resistant gloves will always be worn when handling this equipment as well as long sleeves and eye protection. Equipment will be allowed to sufficiently cool before cleaning or disassembly.

A U-tube manometer is used to monitor pressure differentials across the trays and incorrect valve sequencing can cause vapor backflow into the manometer, leading to condensation or liquid ejection. To control this risk, V6 will always be opened before V7 and closed in reverse to prevent vapor entry.

The reboiler, console and pump system operate using electrical cords and voltage. Accidental water contact or improper grounding may result in electric shock. To avoid risk, connections will be checked to be grounded and cords free from damage. The control panel must stay dry at all times. The cooling water system must also maintain approximately 3L/min to prevent condenser overheating and vapor release. Monitoring the Fl1 flowmeter will be continuous and if the flow drops below 2L/min, the power will be immediately shut off to the reboiler until stable flow is restored.

All flammable liquid waste will be disposed in proper waste containers only.

Figure 3: Safety evaluation chart.

	Yes/No	
Potential electrical hazards?	Y	If yes, identify conditions: Water by electric comp.
Potential mechanical hazards?	N	If yes, identify conditions:
Potential pressure hazards?	Y	If yes, identify conditions: Pressure release line
Potential temperature hazards?	Y	If yes, identify conditions: Reboiler (>90°C), hot product
Hazardous/toxic chemicals involved?	Y	If yes, hazards involved: 200proof Ethanol
Gloves required?	Y	If yes, specify type: heat resistant gloves, nitrile
MSDS reviewed by all team members?	Y	List MSDS sheets reviewed: CAS #64-17-5
Process Parameters	*Max Expected*	*List location and possible hazards and precautions*
Temperature (ºC)	>90°C	Generated by reboiler
Pressure (psia)	Atm	Should only deal w/ atmospheric pressure
Safety Controls	*yes/no*	*If yes, Provide description*
	Y	heat gloves, nitrile gloves, fume hood, Safety Control (Safe area console)

I have read relevant background material for the Unit Operations Laboratory entitled: "Distillation" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Kendell Johnson	10/14/25
signature of team member	date
[illegible]	10/14/25
signature of team member	date
Basil Hopkins	10/14/25
signature of team member	date